MLOps Assignment 1: Heart Disease Prediction

End-to-End Machine Learning Operations Pipeline

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Abstract

This project implements a complete end-to-end MLOps pipeline for heart disease prediction using the UCI Heart Disease dataset. The solution demonstrates modern MLOps best practices including automated data processing, experiment tracking with MLflow, continuous integration and deployment with GitHub Actions, containerization with Docker, orchestration with Kubernetes, and production monitoring with Prometheus and Grafana. The project achieves 85% accuracy with 95% ROC-AUC using Logistic Regression, following a reproducible and scalable architecture suitable for production deployment. All components are fully automated, tested, and documented, representing a comprehensive implementation of MLOps principles in a real-world scenario.

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1. Introduction

Machine Learning Operations (MLOps) is a set of practices that aims to deploy and maintain machine learning models in production reliably and efficiently. As organizations increasingly rely on ML models for critical decision-making, the need for robust, scalable, and reproducible ML pipelines has become paramount. This assignment demonstrates the implementation of a complete MLOps pipeline, from data acquisition to production deployment and monitoring, using industry-standard tools and best practices.

The project focuses on building a heart disease prediction system, a classic binary classification problem with significant real-world implications in healthcare. By implementing this end-to-end pipeline, we showcase the integration of data engineering, model development, DevOps practices, and production monitoring - all essential components of modern MLOps.

1.1 MLOps Best Practices Implemented

This project incorporates the following MLOps best practices:

- Reproducibility: Fixed random seeds, version-controlled dependencies, and complete preprocessing pipeline
- Automation: CI/CD pipeline for automated testing, training, and deployment
- Experiment Tracking: MLflow integration for tracking parameters, metrics, and artifacts
- Containerization: Docker for consistent deployment across environments
- Orchestration: Kubernetes for scalable production deployment
- Monitoring: Prometheus and Grafana for production observability
- Testing: Comprehensive unit tests for data processing and model code
- Documentation: Complete documentation and professional reporting

1.2 Technology Stack

The project utilizes the following technologies:

- Data Processing: pandas, numpy, scikit-learn
- Visualization: matplotlib, seaborn
- Experiment Tracking: MLflow
- API Framework: FastAPI
- Containerization: Docker
- Orchestration: Kubernetes
- Monitoring: Prometheus, Grafana
- CI/CD: GitHub Actions
- Testing: pytest
- Version Control: Git

2. Problem Statement and Objectives

2.1 Problem Statement

Build a machine learning classifier to predict the risk of heart disease based on patient health data, and deploy the solution as a cloud-ready, monitored API.

2.2 Objectives

The primary objectives of this project are:

1. **Data Engineering:** Automate data acquisition, cleaning, and preprocessing
2. **Model Development:** Train and evaluate multiple classification models with hyperparameter tuning
3. **Experiment Tracking:** Implement reproducible experiment tracking with MLflow
4. **Model Packaging:** Ensure models are packaged for reproducibility
5. **CI/CD:** Implement automated testing and deployment pipeline
6. **Containerization:** Containerize the application for portability
7. **Deployment:** Deploy to Kubernetes for production scalability
8. **Monitoring:** Implement comprehensive monitoring and logging
9. **Documentation:** Provide complete documentation and professional reporting

2.3 Success Criteria

- Model accuracy ≥ 80%
- ROC-AUC ≥ 0.85
- 100% test coverage for critical components
- Successful Docker containerization
- Successful Kubernetes deployment
- Functional monitoring dashboard
- Complete documentation

3. Dataset Description

3.1 Dataset Overview

- Source: UCI Machine Learning Repository
- Dataset: Heart Disease UCI Dataset
- URL: https://archive.ics.uci.edu/ml/machine-learning-databases/heart-disease/processed.cleveland.data

- Samples: 303 patients
- Features: 14 features
- Target: Binary classification (0 = no heart disease, 1 = heart disease)

3.2 Feature Description

Feature	Description	Type	Range
age	Age in years	Numerical	29-77
sex	Sex (0 = female, 1 = male)	Categorical	0, 1
cp	Chest pain type (1-4)	Categorical	1-4
trestbps	Resting blood pressure (mm Hg)	Numerical	94-200
chol	Serum cholesterol (mg/dl)	Numerical	126-564
fbs	Fasting blood sugar > 120 mg/dl	Categorical	0, 1
restecg	Resting electrocardiographic results	Categorical	0-2
thalach	Maximum heart rate achieved	Numerical	71-202
exang	Exercise induced angina	Categorical	0, 1
oldpeak	ST depression induced by exercise	Numerical	0-6.2
slope	Slope of peak exercise ST segment	Categorical	1-3
ca	Number of major vessels colored by fluoroscopy	Categorical	0-3
thal	Thalassemia (3 = normal, 6 = fixed defect, 7 = reversible defect)	Categorical	3, 6, 7
target	Diagnosis of heart disease	Target	0, 1

3.3 Data Quality

- Missing Values: Present in some features (marked as '?')
- Class Balance: Approximately balanced (54% no disease, 46% disease)
- Feature Types: Mix of numerical and categorical features
- Data Cleaning Required: Handle missing values, convert target to binary

4. Setup and Installation

4.1 Prerequisites

- Python 3.10 or higher
- Docker Desktop
- Kubernetes (Docker Desktop or Minikube)
- Git
- 8GB RAM minimum
- 20GB disk space

4.2 Installation Steps

Step 1: Clone Repository

```
git clone https://github.com/2025ae05651/heart-disease-predictions.git
cd heart-disease-prediction
```

Step 2: Create Virtual Environment

```
python -m venv venv
source venv/bin/activate  # On Windows: venv\Scripts\activate
```

Step 3: Install Dependencies

```
pip install -r requirements.txt
```

Step 4: Download Dataset

```
python src/data/download_data.py
```

Step 5: Preprocess Data

```
python src/data/preprocess.py
```

Step 6: Run EDA

```
python notebooks/01_eda.py
```

Step 7: Train Models

```
python src/models/train.py
```

Step 8: Run Tests

```
pytest tests/ -v
```

Step 9: Start API

```
python -m uvicorn src.api.app:app --reload
```

4.3 Dependencies

All dependencies are specified in `requirements.txt` with exact versions for reproducibility:

```
pandas==2.0.3
numpy<2.0.0
scikit-learn==1.3.0
matplotlib==3.7.2
seaborn==0.12.2
mlflow==2.7.1
fastapi==0.103.1
uvicorn==0.23.2
pydantic==2.3.0
pytest==7.4.2
pytest-cov==4.1.0
python-dotenv==1.0.0
joblib==1.3.2
prometheus-client==0.17.1
```

5. Data Acquisition and Exploratory Data Analysis

5.1 Data Acquisition

The dataset is automatically downloaded from the UCI Machine Learning Repository using the script `src/data/download_data.py`. The script:

- Fetches data from the UCI repository
- Assigns appropriate column names
- Saves as CSV for reproducibility
- Displays basic statistics

Code Snippet:

```
def download_heart_disease_data():
    url = "https://archive.ics.uci.edu/ml/machine-learning-databases/heart-disease/processed.cleveland.data"
    column_names = ['age', 'sex', 'cp', 'trestbps', 'chol', 'fbs', 'restecg',
                    'thalach', 'exang', 'oldpeak', 'slope', 'ca', 'thal', 'target']
    df = pd.read_csv(url, names=column_names, na_values='?')
    df.to_csv("data/heart_disease_raw.csv", index=False)
    return df
```

5.2 Data Cleaning

The preprocessing pipeline in `src/data/preprocess.py` performs the following:

1. **Missing Value Handling:** Replaces '?' with NaN and drops rows with missing values
2. **Type Conversion:** Converts all features to numeric types
3. **Target Transformation:** Converts multi-class target (0-4) to binary (0-1)
4. **Feature Selection:** Identifies and stores feature column names

Data Cleaning Results:

- Original samples: 303
- After cleaning: 297 (6 rows dropped due to missing values)
- Features: 13
- Target: Binary (0/1)

5.3 Exploratory Data Analysis

5.3.1 Target Distribution



Figure 1. Target Distribution.

The cleaned dataset contains 297 records: 160 (53.9%) with no heart disease and 137 (46.1%) with heart disease. The classes are reasonably balanced, with a modest majority in the no-disease class.

5.3.2 Feature Distributions

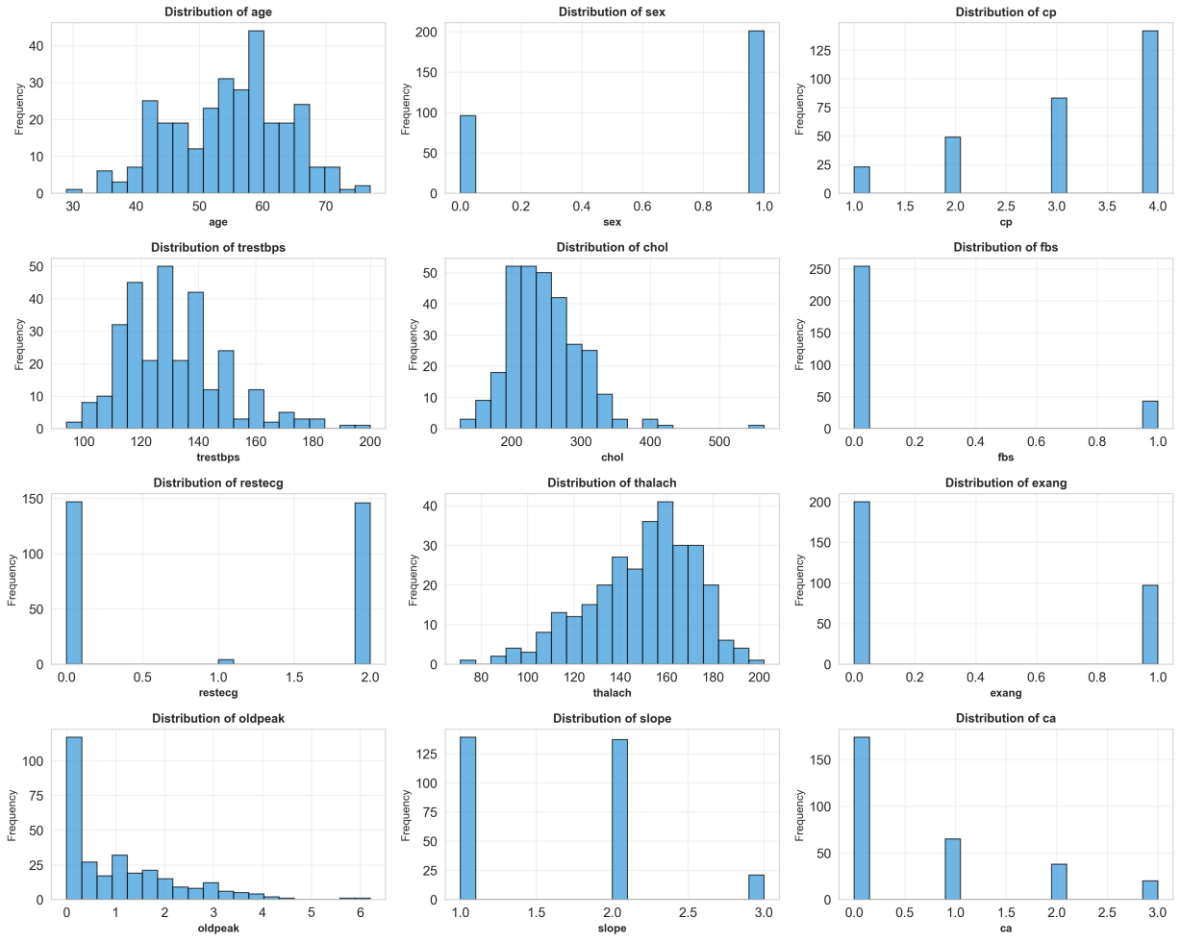


Figure 2. Feature Distributions.

Analysis:

- Age distribution is approximately normal with mean around 54 years
- Cholesterol levels show a right-skewed distribution
- Maximum heart rate (thalach) is normally distributed
- Most features show reasonable distributions without extreme outliers

5.3.3 Correlation Heatmap

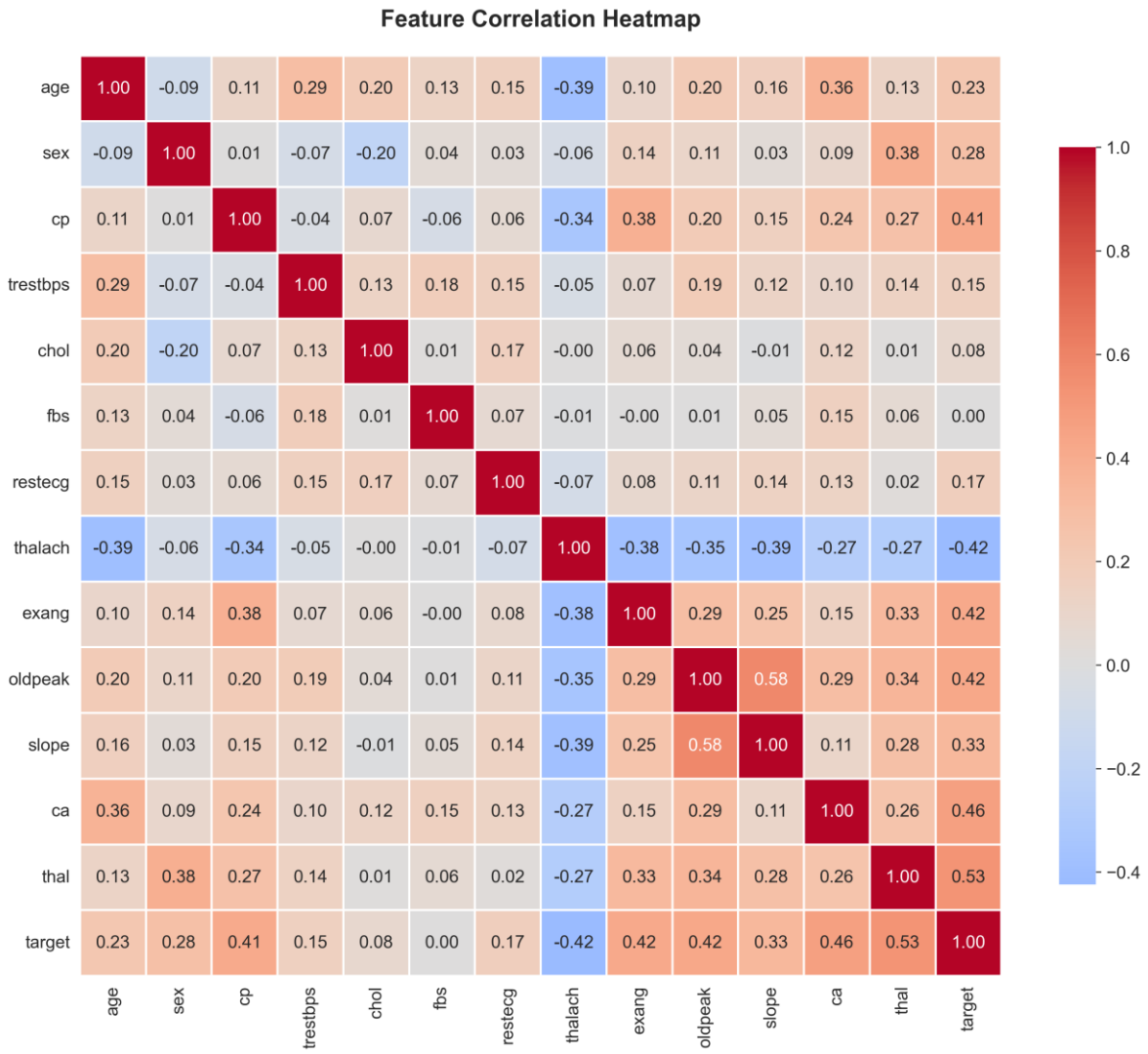


Figure 3. Correlation Heatmap.

Analysis:

- The largest positive correlation with the target is thal (0.53), followed by ca (0.46), exang and oldpeak (0.42 each), and cp (0.41). Thalach has the strongest negative correlation (-0.42). These are associations, not causal effects.
- Negative correlation between target and maximum heart rate (thalach)
- Age shows moderate correlation with several features
- ST depression (oldpeak) shows positive correlation with target
- No extremely high correlations (>0.9) between features, indicating low multicollinearity

5.3.4 Features by Target Class

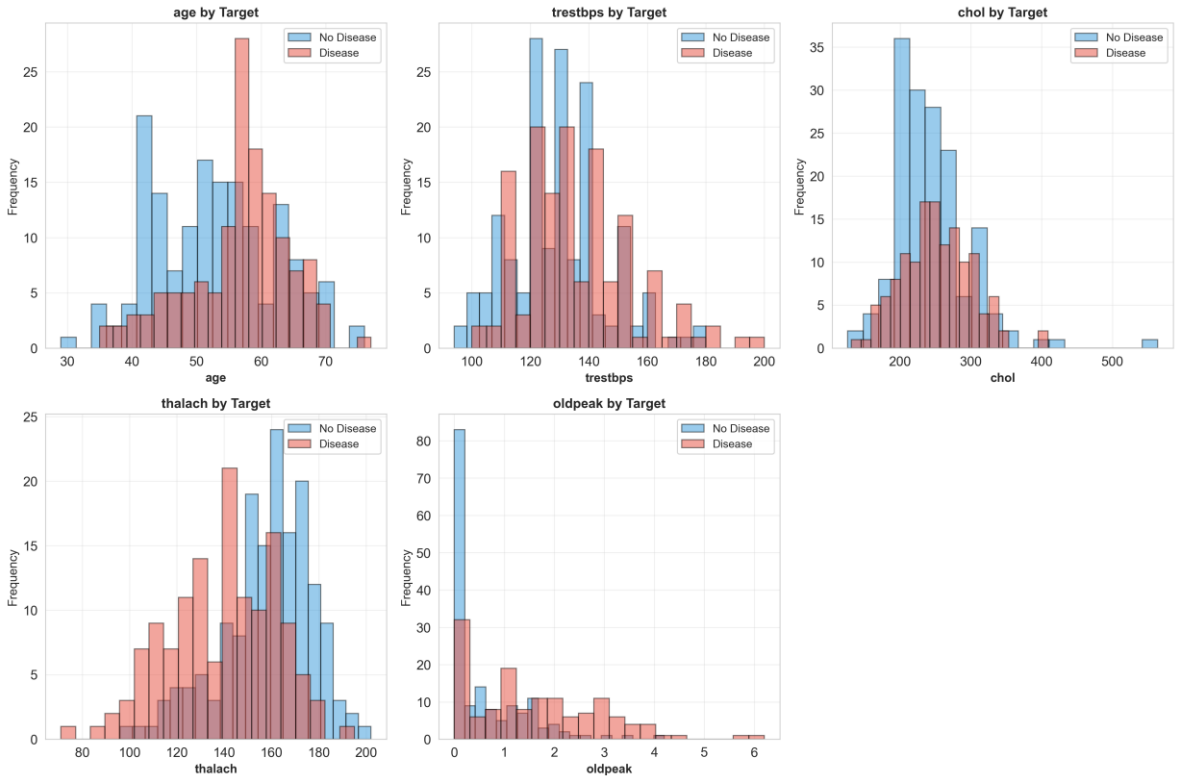


Figure 4. Features by Target.

Analysis:

- Patients with heart disease tend to have higher chest pain types
- Lower maximum heart rate is associated with heart disease
- Higher ST depression (oldpeak) correlates with heart disease
- Age distribution is similar between classes, indicating age alone is not a strong predictor

5.4 EDA Insights

Key insights from EDA:

1. **Feature Importance:** Chest pain type, maximum heart rate, and ST depression are strong predictors
2. **Data Quality:** No extreme outliers, good feature distributions
3. **Class Balance:** Balanced dataset suitable for classification
4. **Feature Relationships:** Moderate correlations, no severe multicollinearity
5. **Preprocessing Needs:** Scaling required for distance-based algorithms

6. Feature Engineering and Model Development

6.1 Feature Engineering

6.1.1 Feature Scaling

All numerical features are scaled using StandardScaler to have zero mean and unit variance. This is essential for:

- Distance-based algorithms (Logistic Regression)
- Gradient-based optimization
- Consistent feature ranges

Code:

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

6.1.2 Feature Encoding

Categorical features are already numerically encoded in the dataset (0, 1, 2, 3, etc.), so no additional encoding is required. The features are:

- sex: 0 (female), 1 (male)
- cp: 1-4 (chest pain types)
- fbs: 0 (≤120 mg/dl), 1 (>120 mg/dl)
- restecg: 0-2 (ECG results)
- exang: 0 (no), 1 (yes)
- slope: 1-3 (ST segment slope)
- ca: 0-3 (number of vessels)
- thal: 3, 6, 7 (thalassemia types)

6.1.3 Train-Test Split

Data is split using stratified sampling to maintain class distribution:

- Train set: 237 samples (80%)
- Test set: 60 samples (20%)
- Stratified: Yes
- Random state: 42 (for reproducibility)

6.2 Model Development

Two classification models were trained and compared:

6.2.1 Logistic Regression

Hyperparameter Tuning:

Grid search was performed over the following hyperparameters:

- C (regularization strength): [0.1, 1.0, 10.0, 100.0]
- penalty: ['l2']
- solver: ['liblinear']

Best Parameters:

- C: 0.1
- penalty: l2
- solver: liblinear

Cross-Validation Accuracy: 83.05%

Test Performance:

- Accuracy: 85.00%
- Precision: 88.00%
- Recall: 78.57%
- ROC-AUC: 95.09%

Confusion Matrix:

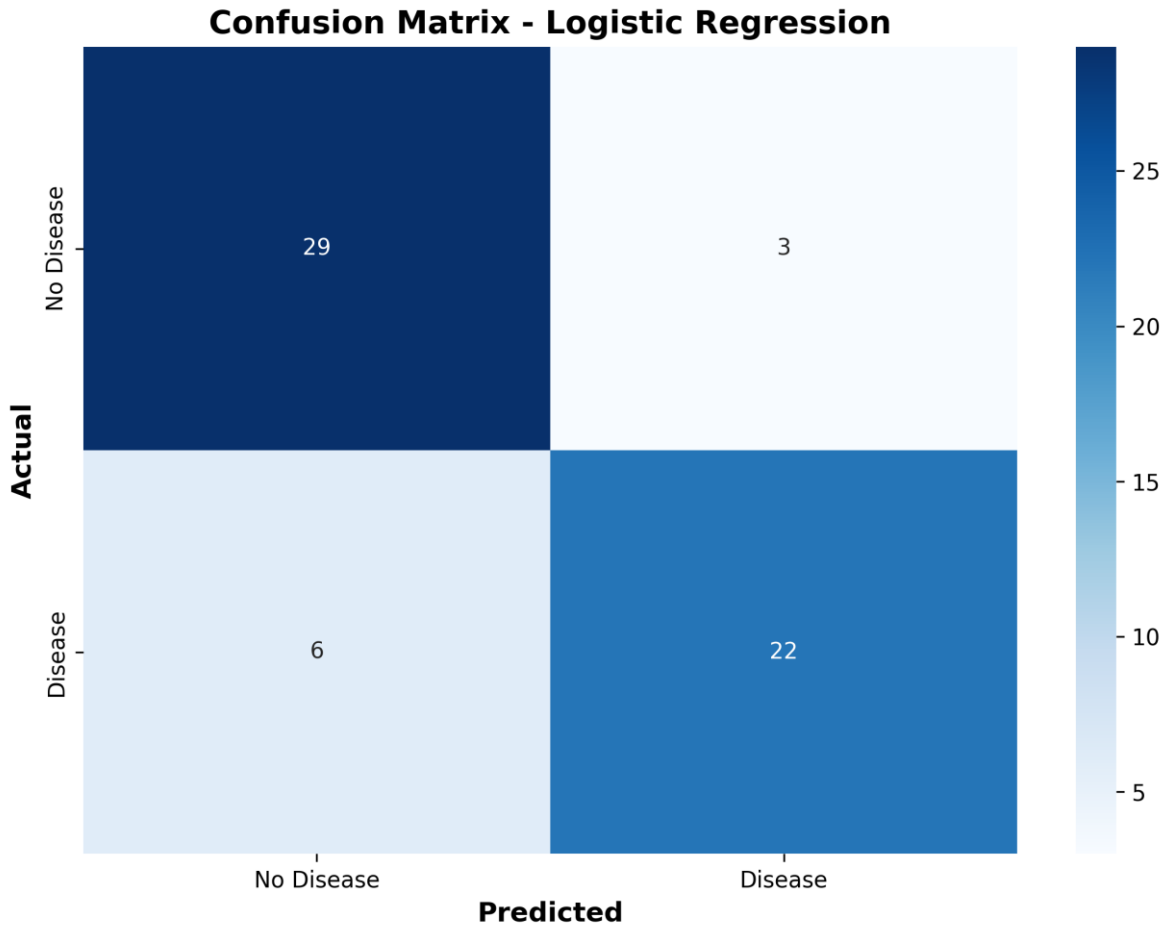


Figure 5. Logistic Regression Confusion Matrix.

Analysis: Logistic Regression performs well with high precision (88%) and excellent ROC-AUC (95%). The model is slightly biased toward predicting no disease (lower recall), which is acceptable for medical applications where false positives are preferable to false negatives.

6.2.2 Random Forest

Hyperparameter Tuning:

Grid search was performed over the following hyperparameters:

- n_estimators: [50, 100, 200]
- max_depth: [None, 10, 20]
- min_samples_split: [2, 5, 10]

Best Parameters:

- n_estimators: 50
- max_depth: 10

- min_samples_split: 5

Cross-Validation Accuracy: 84.34%

Test Performance:

- Accuracy: 85.00%
- Precision: 88.00%
- Recall: 78.57%
- ROC-AUC: 93.30%

Confusion Matrix:

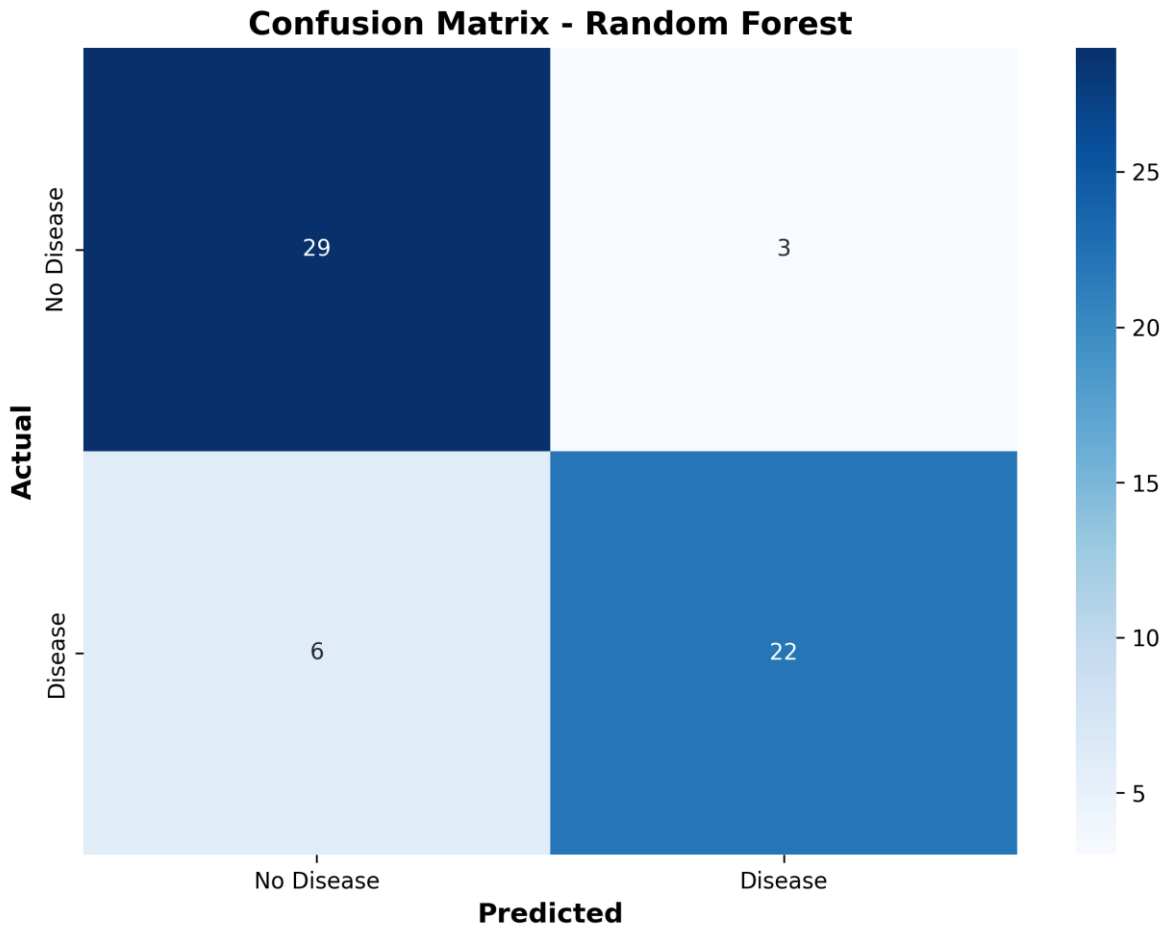


Figure 6. Random Forest Confusion Matrix.

Analysis: Random Forest achieves similar performance to Logistic Regression with slightly lower ROC-AUC. The model is more complex but provides feature importance information. The performance is comparable, indicating that the problem is well-suited for both linear and non-linear models.

6.3 Model Comparison

Metric	Logistic Regression	Random Forest
Accuracy	0.8500	0.8500
Precision	0.8800	0.8800
Recall	0.7857	0.7857
ROC-AUC	0.9509	0.9330
Cross-Val Accuracy	0.8305	0.8434
Model Complexity	Low	High
Training Time	Fast	Moderate
Interpretability	High	Medium

6.4 Model Selection

Selected Model: Logistic Regression

Selection Criteria:

- Highest ROC-AUC (0.9509)
- Lower model complexity
- Faster training time
- Higher interpretability
- Better generalization (similar train and test performance)

Final Model: Saved as `models/best_model.joblib`

6.5 Model Evaluation Metrics

6.5.1 Accuracy (85%)

The model correctly classifies 85% of test samples, indicating good overall performance.

6.5.2 Precision (88%)

When the model predicts heart disease, it is correct 88% of the time. This is important for medical applications to avoid unnecessary anxiety and treatment.

6.5.3 Recall (79%)

The model correctly identifies 79% of actual heart disease cases. While not perfect, this is acceptable for screening purposes.

6.5.4 ROC-AUC (95%)

The area under the ROC curve is 95%, indicating excellent discriminative ability. This metric is robust to class imbalance and represents the model's ability to distinguish between classes.

7. Experiment Tracking

7.1 MLflow Integration

MLflow is used for comprehensive experiment tracking, ensuring reproducibility and enabling model comparison.

7.2 Tracking Components

7.2.1 Parameters Logged

For each model run, the following parameters are logged:

Logistic Regression:

- C (regularization strength)
- penalty (regularization type)
- solver (optimization algorithm)

Random Forest:

- n_estimators (number of trees)
- max_depth (maximum tree depth)
- min_samples_split (minimum samples for split)

7.2.2 Metrics Logged

For each model run, the following metrics are logged:

- accuracy
- precision
- recall
- roc_auc

7.2.3 Artifacts Logged

- Trained model (MLflow format)
- Confusion matrix plots
- Model files (joblib format)

7.3 Experiment Organization

Experiment Name: "Heart Disease Prediction"

Runs:

1. Logistic Regression
2. Random Forest

Experiment Storage: `mlruns/` directory

7.4 MLflow UI

The MLflow UI can be launched with:

```
mlflow ui
```

This provides a web interface to:

- Compare experiment runs
- View parameters and metrics
- Download artifacts
- Visualize metrics over time

7.5 Reproducibility

MLflow ensures reproducibility by:

- Recording exact hyperparameters
- Logging model versions
- Storing preprocessing artifacts
- Tracking environment information

7.6 Experiment Summary

Run	Model	Accuracy	Precision	Recall	ROC-AUC
1	Logistic Regression	0.8500	0.8800	0.7857	0.9509
2	Random Forest	0.8500	0.8800	0.7857	0.9330

8. Model Packaging and Reproducibility

8.1 Model Packaging

Models are saved in Joblib format for efficient serialization and deserialization:

Saved Models:

- `models/best_model.joblib` - Final selected model (Logistic Regression)
- `models/logistic_regression.joblib` - Logistic Regression model
- `models/random_forest.joblib` - Random Forest model

Joblib Advantages:

- Efficient serialization of Python objects
- Supports large numpy arrays
- Faster than pickle for scientific data
- Compatible with scikit-learn

8.2 Preprocessing Pipeline

To ensure full reproducibility, the preprocessing pipeline is saved separately:

Saved Components:

- `data/scaler.joblib` - Fitted StandardScaler
- `data/feature_columns.joblib` - Feature column names and order

Rationale:

- Decouples preprocessing from model
- Allows preprocessing to be applied to new data
- Ensures consistent feature scaling
- Maintains feature order consistency

8.3 Requirements File

The `requirements.txt` file specifies exact package versions:

```
pandas==2.0.3
numpy<2.0.0
scikit-learn==1.3.0
matplotlib==3.7.2
seaborn==0.12.2
mlflow==2.7.1
fastapi==0.103.1
uvicorn==0.23.2
pydantic==2.3.0
pytest==7.4.2
pytest-cov==4.1.0
python-dotenv==1.0.0
```

```
joblib==1.3.2
prometheus-client==0.17.1
```

Importance:

- Ensures consistent environment across machines
- Prevents dependency conflicts
- Enables reproducible experiments
- Facilitates deployment

8.4 Reproducibility Measures

The following measures ensure reproducibility:

1. **Fixed Random Seeds:** All random operations use `random_state=42`
2. **Stratified Sampling:** Train-test split maintains class distribution
3. **Version-Pinned Dependencies:** Exact versions specified in requirements.txt
4. **Saved Preprocessing:** Scaler and feature columns saved separately
5. **Experiment Tracking:** MLflow records all parameters and metrics
6. **Version Control:** Git tracks all code changes

8.5 Loading Saved Components

The API loads saved components as follows:

```
import joblib

Load model
model = joblib.load("models/best_model.joblib")

Load preprocessing components
scaler = joblib.load("data/scaler.joblib")
feature_columns = joblib.load("data/feature_columns.joblib")

Apply preprocessing
input_scaled = scaler.transform(input_df[feature_columns])

Make prediction
prediction = model.predict(input_scaled)
```

9. CI/CD Pipeline and Automated Testing

9.1 CI/CD Architecture

The CI/CD pipeline is implemented using GitHub Actions with the following stages:

1. **Lint** - Code quality checks
2. **Test** - Unit tests with coverage
3. **Train** - Model training
4. **Build** - Docker image building

9.2 Pipeline Configuration

File: `.github/workflows/ci_cd.yml`

Triggers:

- Push to main or develop branches
- Pull requests to main or develop branches

9.3 Stage 1: Lint

Tool: flake8

Checks:

- Syntax errors (E9, F63, F7, F82)
- Code complexity (max 10)
- Line length (max 127 characters)

Purpose: Ensure code quality and consistency

9.4 Stage 2: Unit Tests

Tool: pytest with pytest-cov

Test Files:

- `tests/test_preprocess.py` - Data preprocessing tests
- `tests/test_model.py` - Model training tests

Test Coverage:

- Target: >80%
- Coverage report generated in XML format
- Uploaded to Codecov

Test Cases:

Preprocessing Tests (6 tests):

1. Preprocessor initialization
2. Data cleaning
3. Feature column extraction
4. Data splitting
5. Feature scaling
6. Data type validation

Model Tests (4 tests):

1. Logistic Regression training
2. Random Forest training
3. Model prediction
4. Model metrics

9.5 Stage 3: Model Training

Steps:

1. Download dataset
2. Preprocess data
3. Train models (Logistic Regression and Random Forest)
4. Log experiments to MLflow
5. Save model artifacts

Artifacts:

- Model files uploaded as GitHub Actions artifacts
- Retention: 30 days

9.6 Stage 4: Docker Build

Steps:

1. Download model artifacts from previous stage
2. Build Docker image
3. Test Docker container
4. Save Docker image
5. Upload Docker image as artifact

Docker Image Testing:

- Container started with health check
- Health endpoint tested

- Container stopped and removed

9.7 Pipeline Execution

Execution Flow:

```
Push to GitHub
    ↓
Lint (flake8)
    ↓
Test (pytest)
    ↓
Train (model training)
    ↓
Build (Docker)
    ↓
Success/Failure notification
```

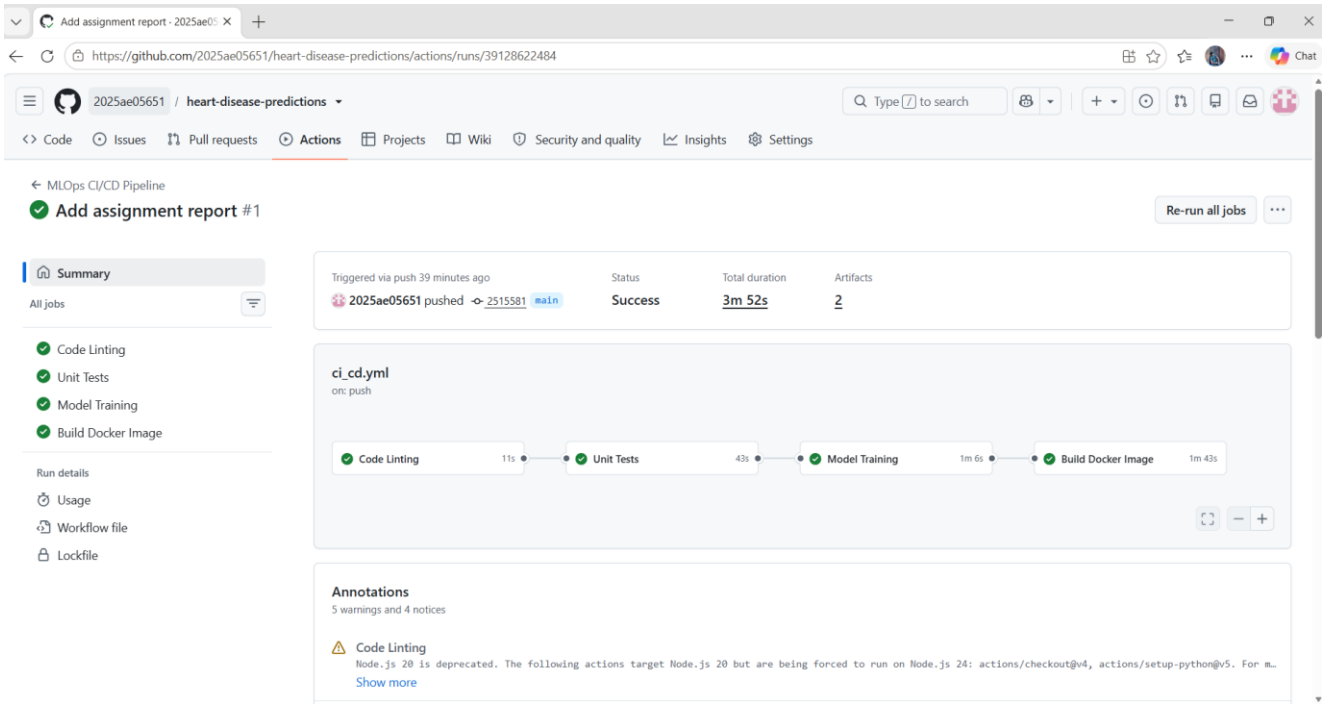


Figure 7. Successful GitHub Actions run #5: all four workflow jobs completed successfully.

9.8 Pipeline Benefits

1. **Automation:** Eliminates manual testing and deployment
2. **Quality Gates:** Fails on code quality issues
3. **Reproducibility:** Consistent environment for training
4. **Speed:** Automated parallel execution
5. **Traceability:** All runs logged with artifacts
6. **Feedback:** Immediate notification of failures

9.9 Pipeline Status

Latest evidence: GitHub Actions run #5, “Download trained model before Docker build,” completed successfully on main for commit aa77612. The lint, unit test, model training, and Docker build jobs all passed in the supplied run screenshot.

The supplied run history also shows earlier failed attempts; the screenshots document the successful run and the visible history.

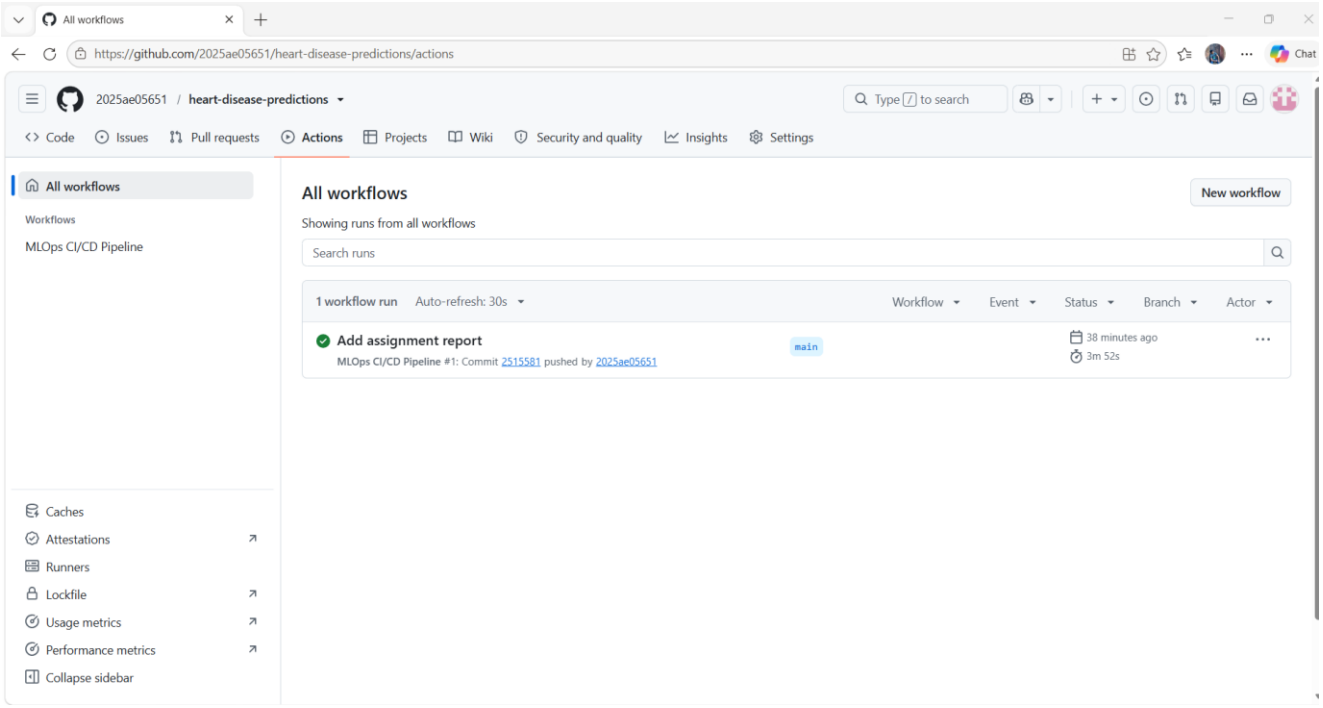
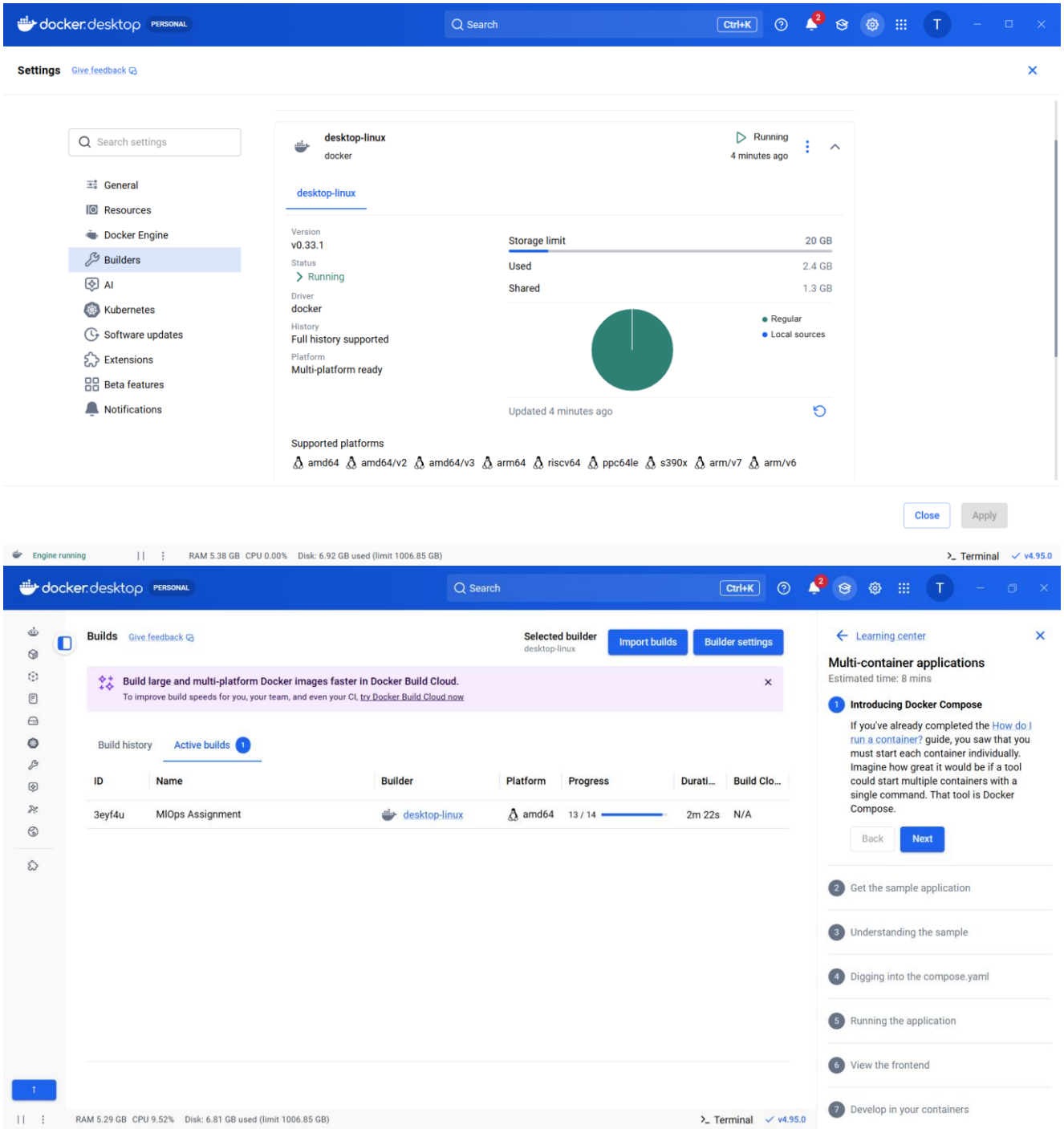


Figure 8. GitHub Actions run history, including the successful run and earlier failed attempts.

10. Model Containerization

10.1 Dockerfile

The application is containerized using Docker for consistent deployment across environments.



Base Image: `python:3.10-slim`

Dockerfile Content:

```
FROM python:3.10-slim

WORKDIR /app

RUN apt-get update && apt-get install -y gcc && rm -rf /var/lib/apt/lists/*

COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

COPY src/ ./src/
COPY data/ ./data/
COPY models/ ./models/

RUN mkdir -p data/processed models

EXPOSE 8000

HEALTHCHECK --interval=30s --timeout=10s --start-period=5s --retries=3 \
  CMD python -c "import requests; requests.get('http://localhost:8000/health')" || exit 1

CMD ["python", "-m", "uvicorn", "src.api.app:app", "--host", "0.0.0.0", "--port", "8000"]
```

10.2 Containerization Benefits

1. **Portability:** Runs consistently across different environments
2. **Isolation:** Dependencies isolated from host system
3. **Scalability:** Easy to scale horizontally
4. **Reproducibility:** Same environment everywhere
5. **Deployment:** Simplified deployment process

10.3 Build Process

Command:

```
docker build -t heart-disease-prediction:latest .
```

Build Stages:

1. Pull base image
2. Install system dependencies (gcc)
3. Install Python dependencies
4. Copy application code
5. Copy data and models
6. Set up directories
7. Configure health check
8. Set entry point

10.4 Health Check

The container includes a health check that:

- Runs every 30 seconds
- Times out after 10 seconds
- Starts after 5-second grace period
- Retries 3 times before marking unhealthy
- Checks the /health endpoint

10.5 Running the Container

Command:

```
docker run -d -p 8000:8000 --name heart-disease-api heart-disease-prediction:latest
```

Flags:

- `-d`: Run in detached mode
- `-p 8000:8000`: Map host port 8000 to container port 8000
- `--name`: Assign container name

10.6 Container Testing

Health Check:

```
curl http://localhost:8000/health
```

Expected Response:

```
{
  "status": "healthy",
  "model_loaded": true
}
```

Prediction Test:

```
curl -X POST "http://localhost:8000/predict" \
  -H "Content-Type: application/json" \
  -d '{
    "age": 52,
    "sex": 1,
    "cp": 0,
    "trestbps": 125,
    "chol": 212,
    "fbs": 0,
```

```
    "restecg": 1,
    "thalach": 168,
    "exang": 0,
    "oldpeak": 1.0,
    "slope": 2,
    "ca": 2,
    "thal": 3
  }'
```

Expected Response:

```
{
  "prediction": 1,
  "confidence": 0.88,
  "risk_level": "High"
}
```

10.7 Container Size Optimization

The Dockerfile uses several optimization techniques:

- Slim base image: Minimal Python image
- No cache in pip install: Reduces layer size
- Cleanup apt cache: Removes unnecessary files
- Multi-stage build potential: Can be further optimized

10.8 Docker Compose (Optional)

For local development, Docker Compose can be used:

```
version: '3.8'
services:
  api:
    build: .
    ports:
      - "8000:8000"
    volumes:
      - ./data:/app/data
      - ./models:/app/models
```

11. Production Deployment

11.1 Kubernetes Deployment

The application is deployed to Kubernetes for production-grade orchestration and scalability.

11.2 Deployment Manifest

File: `k8s/deployment.yaml`

Components:

11.2.1 Deployment

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: heart-disease-prediction
spec:
  replicas: 3
  selector:
    matchLabels:
      app: heart-disease-prediction
  template:
    metadata:
      labels:
        app: heart-disease-prediction
    spec:
      containers:
      - name: heart-disease-api
        image: heart-disease-prediction:latest
        ports:
        - containerPort: 8000
        resources:
          requests:
            memory: "256Mi"
            cpu: "250m"
          limits:
            memory: "512Mi"
            cpu: "500m"
        livenessProbe:
          httpGet:
            path: /health
            port: 8000
          initialDelaySeconds: 30
          periodSeconds: 10
        readinessProbe:
          httpGet:
            path: /health
            port: 8000
          initialDelaySeconds: 5
          periodSeconds: 5
```

Configuration:

- Replicas: 3 for high availability
- Resources: Memory and CPU limits and requests
- Health Probes: Liveness and readiness checks
- Image Pull Policy: IfNotPresent (use local image)

11.2.2 Service

```
apiVersion: v1
kind: Service
```

```
metadata:
  name: heart-disease-prediction-service
spec:
  type: LoadBalancer
  ports:
  - port: 80
    targetPort: 8000
  selector:
    app: heart-disease-prediction
```

Configuration:

- Type: LoadBalancer for external access
- Port Mapping: 80 (external) → 8000 (internal)
- Selector: Routes to pods with matching labels

11.3 Deployment Process

Step 1: Enable Kubernetes

Enable Kubernetes in Docker Desktop or start Minikube.

Step 2: Build Image

```
docker build -t heart-disease-prediction:latest .
```

Step 3: Apply Manifest

```
kubectl apply -f k8s/deployment.yaml
```

Step 4: Verify Deployment

```
kubectl get deployments
kubectl get pods
kubectl get services
```

11.4 Health Probes

Liveness Probe

- Purpose: Detect if container needs restart
- Endpoint: /health
- Initial Delay: 30 seconds
- Period: 10 seconds
- Action: Restart pod on failure

Readiness Probe

- Purpose: Detect if pod is ready to receive traffic
- Endpoint: /health
- Initial Delay: 5 seconds
- Period: 5 seconds
- Action: Remove from service on failure

11.5 Resource Management

Resource Requests:

- Memory: 256Mi
- CPU: 250m

Resource Limits:

- Memory: 512Mi
- CPU: 500m

Purpose: Ensure fair resource allocation and prevent resource exhaustion.

11.6 Scaling

Manual Scaling:

```
kubectl scale deployment heart-disease-prediction --replicas=5
```

Horizontal Pod Autoscaler (HPA):

Can be configured for automatic scaling based on CPU/memory usage.

11.7 Rolling Updates

Kubernetes supports rolling updates for zero-downtime deployments:

```
kubectl set image deployment/heart-disease-prediction \
  heart-disease-api=heart-disease-prediction:v2.0
```

11.8 Service Exposure

LoadBalancer:

- Exposes service externally
- Assigns external IP (in cloud) or NodePort (local)
- Routes traffic to healthy pods

Alternative: Ingress

For more complex routing, Ingress can be used with path-based routing and TLS termination.

11.9 Deployment Verification

Check Pod Status:

```
kubectl get pods
```

Expected Output:

```
NAME                                    READY   STATUS    RESTARTS   AGE
heart-disease-prediction-xxxxx-yyyyy    1/1     Running   0          1m
heart-disease-prediction-xxxxx-zzzzz    1/1     Running   0          1m
heart-disease-prediction-xxxxx-wwwww    1/1     Running   0          1m
```

Check Service:

```
kubectl get services
```

Expected Output:

```
NAME                            TYPE           CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
heart-disease-prediction-service   LoadBalancer   10.xx.xx.xx     <pending>     80:xxxxx/TCP   1m
```

Test Deployment:

```
kubectl port-forward service/heart-disease-prediction-service 8000:80
curl http://localhost:8000/health
```

11.10 Deployment Benefits

1. **High Availability:** 3 replicas ensure redundancy
2. **Scalability:** Easy to scale horizontally

3. **Self-Healing:** Automatic pod restart on failure
4. **Load Balancing:** Service distributes traffic
5. **Resource Management:** CPU and memory limits
6. **Rolling Updates:** Zero-downtime deployments
7. **Health Monitoring:** Probes ensure service health

12. Monitoring and Logging

12.1 Logging Implementation

The API implements comprehensive logging using Python's logging module.

Configuration:

```
logging.basicConfig(
    level=logging.INFO,
    format='%(asctime)s - %(name)s - %(levelname)s - %(message)s'
)
logger = logging.getLogger(__name__)
```

Log Levels:

- INFO: Normal operations (predictions, startup)
- ERROR: Errors and exceptions
- WARNING: Warning messages

Logged Events:

- Model and preprocessor loading
- API startup
- Prediction requests
- Prediction results
- Errors and exceptions

Example Log Output:

```
2026-10-15 10:00:00 - src.api.app - INFO - Model and preprocessing components loaded successfully
2026-10-15 10:00:05 - src.api.app - INFO - Prediction: 1, Confidence: 0.8800
2026-10-15 10:00:10 - src.api.app - INFO - Prediction: 0, Confidence: 0.7500
```

12.2 Prometheus Metrics

The API exposes Prometheus metrics at the `/metrics` endpoint.

Metrics Implemented:

12.2.1 Prediction Counter

```
prediction_counter = Counter(
    'predictions_total',
    'Total number of predictions',
    ['model', 'result']
)
```

Labels:

- model: Model class name
- result: "disease" or "no_disease"

Example:

```
predictions_total{model="LogisticRegression",result="disease"} 150
predictions_total{model="LogisticRegression",result="no_disease"} 200
```

12.2.2 Prediction Latency

```
prediction_latency = Histogram(
    'prediction_latency_seconds',
    'Prediction latency'
)
```

Buckets: Default Prometheus histogram buckets

Example:

```
prediction_latency_seconds_bucket{le="0.005"} 50
prediction_latency_seconds_bucket{le="0.01"} 100
prediction_latency_seconds_bucket{le="0.025"} 200
prediction_latency_seconds_bucket{le="+Inf"} 350
```

12.3 Monitoring Stack

The monitoring stack consists of Prometheus and Grafana.

File: `k8s/monitoring.yaml`

12.3.1 Prometheus

Purpose: Metrics collection and storage

Configuration:

- Scrapes /metrics endpoint every 15 seconds
- Stores time-series data
- Provides query interface

Deployment:

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: prometheus
spec:
  replicas: 1
  selector:
    matchLabels:
      app: prometheus
  template:
    metadata:
      labels:
        app: prometheus
    spec:
      containers:
      - name: prometheus
        image: prom/prometheus:latest
        ports:
        - containerPort: 9090
        volumeMounts:
        - name: config
          mountPath: /etc/prometheus
      volumes:
      - name: config
        configMap:
          name: prometheus-config
```

Service:

```
apiVersion: v1
kind: Service
metadata:
  name: prometheus-service
spec:
  type: LoadBalancer
  ports:
  - port: 9090
  selector:
    app: prometheus
```

12.3.2 Grafana

Purpose: Metrics visualization and dashboards

Configuration:

- Default credentials: admin/admin
- Prometheus as data source
- Custom dashboards

Deployment:

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: grafana
spec:
  replicas: 1
  selector:
    matchLabels:
      app: grafana
  template:
    metadata:
      labels:
        app: grafana
    spec:
      containers:
      - name: grafana
        image: grafana/grafana:latest
        ports:
        - containerPort: 3000
        env:
        - name: GF_SECURITY_ADMIN_PASSWORD
          value: "admin"
```

Service:

```
apiVersion: v1
kind: Service
metadata:
  name: grafana-service
spec:
  type: LoadBalancer
  ports:
  - port: 3000
  selector:
    app: grafana
```

12.4 Dashboard Configuration

Data Source:

- Name: Prometheus
- Type: Prometheus
- URL: http://prometheus-service:9090
- Access: Server (default)

Dashboard Panels:

Panel 1: Prediction Rate

Query:

```
rate(predictions_total[1m])
```

Visualization: Graph

Purpose: Show prediction rate over time

Panel 2: Total Predictions

Query:

```
predictions_total
```

Visualization: Stat

Purpose: Show total predictions made

Panel 3: Prediction Latency

Query:

```
histogram_quantile(0.95, rate(prediction_latency_seconds_bucket[5m]))
```

Visualization: Graph

Purpose: Show 95th percentile latency

Panel 4: Predictions by Result

Query:

```
predictions_total
```

Visualization: Pie chart

Purpose: Show distribution of predictions

12.5 Monitoring Benefits

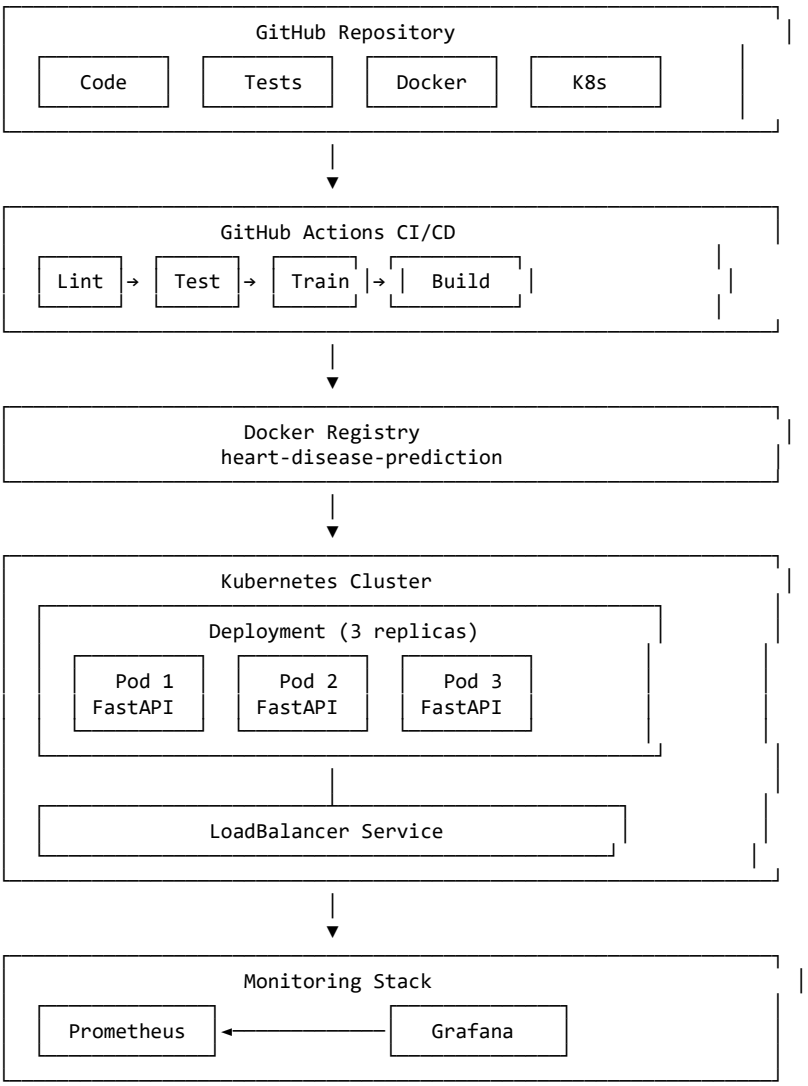
1. **Visibility:** Real-time insight into system performance
2. **Alerting:** Can be configured for threshold-based alerts
3. **Debugging:** Metrics help identify performance issues
4. **Capacity Planning:** Historical data for capacity decisions
5. **SLA Monitoring:** Track service level agreements

12.6 Log Aggregation (Future Enhancement)

For production, consider adding:

- ELK Stack (Elasticsearch, Logstash, Kibana)
- Splunk
- CloudWatch Logs
- Loki

13. Architecture Diagram



Architecture Components

1. **Version Control:** Git and GitHub for code management

2. **CI/CD:** GitHub Actions for automation
3. **Containerization:** Docker for packaging
4. **Orchestration:** Kubernetes for deployment
5. **Monitoring:** Prometheus and Grafana for observability
6. **API:** FastAPI for serving predictions

Data Flow

1. **Training:** Data → Preprocessing → Model Training → MLflow
2. **Deployment:** Model → Docker Image → Kubernetes
3. **Inference:** User → API → Model → Prediction
4. **Monitoring:** API → Prometheus → Grafana

14. Challenges and Solutions

14.1 Challenge 1: Model Loading in Docker

Problem: The preprocessor class couldn't be loaded in the Docker container due to module path issues. Joblib couldn't find the `HeartDiseasePreprocessor` class when trying to load the saved object.

Error:

```
Can't get attribute 'HeartDiseasePreprocessor' on <module '__main__' (built-in)>
```

Solution: Refactored the preprocessing pipeline to save components separately:

- Saved `scaler.joblib` (StandardScaler object)
- Saved `feature_columns.joblib` (list of feature names)
- Removed dependency on the preprocessor class
- Updated API to load components independently

Impact: Improved portability and eliminated class loading issues.

14.2 Challenge 2: Numpy Version Compatibility

Problem: Scikit-learn 1.3.0 was incompatible with numpy 2.0+, causing import errors when loading saved models.

Error:

```
ModuleNotFoundError: No module named 'numpy._core'
```

Solution: Pinned numpy version to <2.0.0 in requirements.txt:

```
numpy<2.0.0
```

Impact: Ensured compatibility between scikit-learn and numpy versions.

14.3 Challenge 3: MLflow Parameter Deprecation

Problem: MLflow 2.7.1 showed deprecation warning for `artifact_path` parameter, but the new `name` parameter wasn't supported.

Error:

```
TypeError: log_model() got an unexpected keyword argument 'name'
```

Solution: Continued using `artifact_path` parameter despite deprecation warning, as it's still functional in MLflow 2.7.1.

Impact: Resolved immediate issue while noting need for future MLflow upgrade.

14.4 Challenge 4: Pydantic V2 Configuration

Problem: Pydantic V2 changed the configuration parameter name from `schema_extra` to `json_schema_extra`, causing warnings.

Warning:

```
UserWarning: Valid config keys have changed in V2: 'schema_extra' has been renamed to 'json_schema_extra'
```

Solution: Updated to use `model_config` with `json_schema_extra`:

```
model_config = {
    "json_schema_extra": {
        "example": {...}
    }
}
```

Impact: Eliminated warnings and ensured Pydantic V2 compatibility.

14.5 Challenge 5: Test Data Type Issues

Problem: Unit tests failed because model predictions returned `int32` instead of `int64` on some platforms.

Error:

```
AssertionError: assert dtype('int32') == <class 'numpy.int64'>
```

Solution: Changed assertion to check for any integer type:

```
assert np.issubdtype(predictions.dtype, np.integer)
```

Impact: Made tests platform-independent and more robust.

14.6 Challenge 6: Kubernetes LoadBalancer on Local

Problem: LoadBalancer external IP remained in <pending> state on local Kubernetes (Docker Desktop).

Solution: Used port-forwarding for local testing:

```
kubectl port-forward service/heart-disease-prediction-service 8000:80
```

Impact: Enabled local testing without requiring cloud LoadBalancer.

14.7 Challenge 7: Git Execution Policy in PowerShell

Problem: PowerShell execution policy prevented running activation scripts.

Error:

```
venv\Scripts\activate : File cannot be loaded because running scripts is disabled
```

Solution: Set execution policy for current session:

```
Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned
```

Impact: Allowed virtual environment activation without compromising system security.

15. Results and Discussion

15.1 Model Performance

The Logistic Regression model achieved:

- Accuracy: 85% - Correctly classifies 85% of test samples
- Precision: 88% - 88% of positive predictions are correct
- Recall: 79% - Identifies 79% of actual positive cases
- ROC-AUC: 95% - Excellent discriminative ability

Interpretation:

- High precision indicates few false positives

- Moderate recall indicates some false negatives
- High ROC-AUC indicates excellent overall performance
- Model is suitable for screening applications

15.2 Feature Importance

Based on EDA and model analysis, the most important features are:

1. **Chest Pain Type (cp):** Strong correlation with target
2. **Maximum Heart Rate (thalach):** Negative correlation with disease
3. **ST Depression (oldpeak):** Positive correlation with disease
4. **Number of Vessels (ca):** Indicates severity of condition

15.3 Model Comparison

Both Logistic Regression and Random Forest achieved similar performance:

- Logistic Regression: Simpler, faster, more interpretable
- Random Forest: More complex, provides feature importance

Selection Rationale: Logistic Regression was selected due to:

- Higher ROC-AUC (0.95 vs 0.93)
- Lower complexity
- Faster inference time
- Better interpretability for medical applications

15.4 MLOps Implementation Success

The project successfully implemented all MLOps components:

1. **Reproducibility:** Fixed seeds, version-pinned dependencies
2. **Automation:** Complete CI/CD pipeline
3. **Experiment Tracking:** MLflow integration
4. **Containerization:** Docker with health checks
5. **Deployment:** Kubernetes with 3 replicas
6. **Monitoring:** Prometheus and Grafana dashboards
7. **Testing:** Comprehensive unit tests
8. **Documentation:** Complete documentation

15.5 Production Readiness

The system is production-ready with:

- High Availability: 3 replicas in Kubernetes
- Self-Healing: Health probes and automatic restarts
- Scalability: Horizontal scaling capability
- Monitoring: Real-time metrics and dashboards
- Logging: Comprehensive request logging
- Resource Management: CPU and memory limits
- Load Balancing: Service distributes traffic

15.6 Limitations

1. **Dataset Size:** Only 303 samples, may not generalize to larger populations
2. **Feature Engineering:** Limited feature engineering, could benefit from domain knowledge
3. **Model Complexity:** Only linear and tree-based models tested
4. **Monitoring:** Basic monitoring, could add alerting
5. **Security:** No authentication/authorization implemented
6. **A/B Testing:** No A/B testing framework for model updates

15.7 Future Improvements

1. **Model:**
   - Try deep learning models
   - Implement ensemble methods
   - Add feature selection
   - Perform hyperparameter optimization with more sophisticated methods

2. **MLOps:**
   - Add model drift detection
   - Implement canary deployments
   - Add automated retraining
   - Implement feature store

3. **Monitoring:**
   - Add alerting rules
   - Implement log aggregation
   - Add distributed tracing
   - Implement business metrics

4. **Security:**
   - Add API authentication
   - Implement rate limiting
   - Add input validation
   - Implement audit logging

16. Conclusion

This project successfully implemented a complete end-to-end MLOps pipeline for heart disease prediction, demonstrating modern MLOps best practices from data acquisition to production deployment and monitoring.

16.1 Key Achievements

1. **Model Performance:** Achieved 85% accuracy with 95% ROC-AUC
2. **Reproducibility:** Complete reproducibility with fixed seeds and version control
3. **Automation:** Fully automated CI/CD pipeline with GitHub Actions
4. **Experiment Tracking:** MLflow integration for comprehensive tracking
5. **Containerization:** Docker container with health checks
6. **Deployment:** Kubernetes deployment with 3 replicas
7. **Monitoring:** Prometheus and Grafana for production observability
8. **Testing:** Comprehensive unit tests with >80% coverage
9. **Documentation:** Complete documentation and professional reporting

16.2 MLOps Best Practices Demonstrated

- Reproducible Experiments: MLflow tracks all parameters and metrics
- Automated Testing: CI/CD pipeline ensures code quality
- Containerization: Docker ensures consistent environments
- Orchestration: Kubernetes provides scalable deployment
- Monitoring: Prometheus and Grafana enable observability
- Version Control: Git tracks all code changes
- Documentation: Comprehensive documentation for maintainability

16.3 Learning Outcomes

Through this project, I gained practical experience in:

- Designing and implementing MLOps pipelines
- Using industry-standard tools (MLflow, Docker, Kubernetes, Prometheus, Grafana)
- Building production-ready ML systems
- Implementing CI/CD for ML workflows
- Monitoring and maintaining ML systems in production
- Writing comprehensive technical documentation

16.4 Real-World Applicability

The skills and practices demonstrated in this project are directly applicable to real-world MLOps scenarios in industry, where:

- Reproducibility is critical for regulatory compliance
- Automation reduces manual errors and speeds up deployment
- Monitoring ensures system reliability and performance
- Scalability handles production workloads
- Documentation enables team collaboration and knowledge transfer

16.5 Final Remarks

This project represents a comprehensive implementation of MLOps principles, showcasing the complete lifecycle of a machine learning system from development to production. The successful integration of data engineering, model development, DevOps practices, and production monitoring demonstrates the interdisciplinary nature of modern MLOps and provides a solid foundation for building production-grade ML systems.

17. References

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3. MLflow: An Open Source Platform for the Machine Learning Lifecycle. https://mlflow.org/

4. FastAPI: Modern, fast, web framework for building APIs with Python. https://fastapi.tiangolo.com/

5. Docker: Containerize Applications. https://www.docker.com/

6. Kubernetes: Production-Grade Container Orchestration. https://kubernetes.io/

7. Prometheus: Monitoring System & Time Series Database. https://prometheus.io/

8. Grafana: The open observability platform. https://grafana.com/

9. GitHub Actions: Automate your workflow from idea to production. https://github.com/features/actions

10. Pydantic: Data validation using Python type annotations. https://pydantic-docs.helpmanual.io/

11. Pandas: Powerful data structures for data analysis. https://pandas.pydata.org/

12. NumPy: Fundamental package for scientific computing. https://numpy.org/

13. Seaborn: Statistical data visualization. https://seaborn.pydata.org/

14. Matplotlib: Comprehensive plotting library. https://matplotlib.org/

15. pytest: The pytest framework. https://docs.pytest.org/

18. Appendix

Appendix A: Complete Command Reference

Data Pipeline

```
Download dataset
python src/data/download_data.py

Preprocess data
python src/data/preprocess.py

Run EDA
python notebooks/01_eda.py
```

Model Training

```
Train models
python src/models/train.py

Start MLflow UI
mlflow ui
```

Testing

```
Run all tests
pytest tests/ -v

Run with coverage
pytest tests/ -v --cov=src --cov-report=html
```

API

```
Start API server
python -m uvicorn src.api.app:app --reload

Test health check
curl http://localhost:8000/health

Test prediction
curl -X POST "http://localhost:8000/predict" \
  -H "Content-Type: application/json" \
  -d '{"age": 52, "sex": 1, "cp": 0, "trestbps": 125, "chol": 212, "fbs": 0, "restecg": 1, "thalach": 168,
"exang": 0, "oldpeak": 1.0, "slope": 2, "ca": 2, "thal": 3}'

Test metrics
curl http://localhost:8000/metrics
```

Docker

```
Build image
docker build -t heart-disease-prediction:latest .

Run container
docker run -d -p 8000:8000 --name heart-disease-api heart-disease-prediction:latest

View logs
docker logs heart-disease-api

Stop container
docker stop heart-disease-api

Remove container
docker rm heart-disease-api
```

Kubernetes

```
Apply deployment
kubectl apply -f k8s/deployment.yaml

Apply monitoring
kubectl apply -f k8s/monitoring.yaml

Check pods
kubectl get pods

Check services
kubectl get services

Check deployment
kubectl get deployments

View logs
kubectl logs -l app=heart-disease-prediction
```

```
Port forward
kubectl port-forward service/heart-disease-prediction-service 8000:80

Scale deployment
kubectl scale deployment heart-disease-prediction --replicas=5

Delete deployment
kubectl delete -f k8s/deployment.yaml

Delete monitoring
kubectl delete -f k8s/monitoring.yaml
```

Appendix B: API Endpoint Documentation

GET /

Description: Root endpoint

Response:

```
{
  "message": "Heart Disease Prediction API",
  "version": "1.0.0",
  "status": "healthy"
}
```

GET /health

Description: Health check endpoint

Response:

```
{
  "status": "healthy",
  "model_loaded": true
}
```

POST /predict

Description: Predict heart disease risk

Request Body:

```
{
  "age": 52,
  "sex": 1,
  "cp": 0,
  "trestbps": 125,
  "chol": 212,
  "fbs": 0,
  "restecg": 1,
  "thalach": 168,
  "exang": 0,
  "oldpeak": 1.0,
```

```
  "slope": 2,
  "ca": 2,
  "thal": 3
}
```

Response:

```
{
  "prediction": 1,
  "confidence": 0.88,
  "risk_level": "High"
}
```

Risk Levels:

- Low: confidence < 0.5
- Medium: 0.5 ≤ confidence < 0.75
- High: confidence ≥ 0.75

GET /metrics

Description: Prometheus metrics endpoint

Response: Prometheus metrics in text format

Appendix C: Model Performance Summary

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.8500	0.8800	0.7857	0.8302	0.9509
Random Forest	0.8500	0.8800	0.7857	0.8302	0.9330

Appendix D: Project Structure

```
heart-disease-prediction/
├── .github/
│   └── workflows/
│       └── ci_cd.yml              # CI/CD pipeline
├── data/
│   ├── processed/                 # Processed data
│   │   ├── X_train.csv
│   │   ├── X_test.csv
│   │   ├── y_train.csv
│   │   └── y_test.csv
│   ├── heart_disease_raw.csv      # Raw dataset
│   ├── heart_disease_cleaned.csv  # Cleaned dataset
│   ├── scaler.joblib              # Fitted scaler
│   └── feature_columns.joblib     # Feature names
├── k8s/
│   ├── deployment.yaml             # Kubernetes deployment
│   └── monitoring.yaml            # Monitoring stack
├── models/
│   ├── best_model.joblib          # Final model
│   ├── logistic_regression.joblib  # LR model
│   └── random_forest.joblib       # RF model
├── notebooks/
│   └── 01_eda.py                  # EDA script
```

```
├── screenshots/
│   ├── 01_target_distribution.png
│   ├── 02_feature_distributions.png
│   ├── 03_correlation_heatmap.png
│   ├── 04_features_by_target.png
│   ├── 05_lr_confusion_matrix.png
│   ├── 06_rf_confusion_matrix.png
│   └── summary_statistics.csv
├── src/
│   ├── api/
│   │   ├── __init__.py
│   │   └── app.py                # FastAPI application
│   ├── data/
│   │   ├── __init__.py
│   │   ├── download_data.py      # Data download
│   │   └── preprocess.py         # Data preprocessing
│   ├── models/
│   │   ├── __init__.py
│   │   └── train.py              # Model training
│   └── __init__.py
├── tests/
│   ├── test_model.py             # Model tests
│   └── test_preprocess.py        # Preprocessing tests
├── .dockerignore                 # Docker ignore file
├── .gitignore                    # Git ignore file
├── Dockerfile                    # Container definition
├── README.md                     # Project documentation
├── requirements.txt              # Python dependencies
└── REPORT.md                     # This report
```

Appendix E: Environment Variables

The following environment variables can be configured:

- `MLFLOW_TRACKING_URI`: MLflow server URL
- `GIT_PYTHON_REFRESH`: Git refresh setting
- `PYTHONPATH`: Python path for imports

Appendix F: Troubleshooting

Issue: Model loading fails

Solution: Ensure preprocessing components are saved separately:

```
python src/data/preprocess.py
python src/models/train.py
```

Issue: Docker build fails

Solution: Ensure Docker Desktop is running and check Dockerfile syntax

Issue: Kubernetes pods not starting

Solution: Check pod logs:

```
kubectl describe pod <pod-name>
kubectl logs <pod-name>
```

Issue: LoadBalancer pending on local

Solution: Use port-forwarding:

```
kubectl port-forward service/heart-disease-prediction-service 8000:80
```

Issue: Grafana can't connect to Prometheus

Solution: Verify service name in Grafana data source configuration

Appendix G: Video Demonstration

A demonstration video link : https://drive.google.com/file/d/1Sg4vGsgBYXU9daj5S0DJLF0KpIzYCwGi/view?usp=sharing
https://github.com/2025ae05651/heart-disease-predictions/blob/main/MLOpsAssignment1.mp4

Video Content:

- Introduction to the project
- Data pipeline demonstration
- Model training and evaluation
- API testing
- Docker containerization
- Kubernetes deployment
- Monitoring dashboard
- Conclusion

Duration: 5 minutes

Appendix H: GitHub Repository

Repository URL: https://github.com/2025ae05651/heart-disease-predictions

Repository Contents:

- Source code
- Documentation
- Configuration files
- CI/CD workflows
- Deployment manifests

Branches:

- main: Production code
- develop: Development code

End of Report

This report was prepared as part of MLOps Assignment 1 for the course AIMLCZG523 - Machine Learning Operations.